Beyond the Code:
Human skills in the AI-era

Gabriella Frank Ferm

Systemutveckling .NET

20260302

Abstract

Denna rapport undersöker hur AI:s allt större roll i mjukvaruutveckling förändrar rollen som utvecklare, och vilka mänskliga färdigheter som blir avgörande när AI genererar en större del av koden. Genom att sammanställa forskning och branschperspektiv blir det tydligt att utvecklare rör sig mot mer övergripande ansvar – som arkitektur, kvalitet, problemlösning och etiska avvägningar – medan AI tar hand om det mer mekaniska kodandet. Resultaten visar att AI kan ta över stora delar av kodproduktionen, men inte ersätta utvecklare; i stället förändras hur deras kompetens kommer till användning. I framtiden blir utvecklarens styrkor som kreativitet, omdöme och förmågan att se helheten ännu viktigare.

This report looks at how AI's increasing presence in software development is changing the developer role and which human skills matter most when AI writes most of the code. By reviewing both research and industry views, it becomes clear that developers are moving toward broader tasks like architecture, quality, problem-solving and ethical considerations while AI takes care of routine coding. The findings show that AI can handle much of the code writing, but it does not replace developers. Instead, it changes how their skills are used. In the future, qualities like creativity, good judgment, and seeing the bigger picture will be even more valuable for developers.

Abbreviations and Concepts

AI – Artificial Intelligence
Technology that enables automated analysis, generation and problem solving – for example by producing code.

LLM - Large Language Model
A type of AI model trained on large amounts of text, capable of generating, explaining and modifying code and text.

AI-assisted development
The use of AI tools that generate, suggest or modify code and support the development process, for example through debugging or test generation.

AI-generated code
Code that has been fully or partially produced by an AI system, such as GitHub Copilot or ChatGPT.

Human skills
In this report, this refers to skills such as architectural thinking, critical reasoning, creativity, collaboration, communication and ethical judgement.

Coding knowledge
The ability to read, understand, review and evaluate code, regardless of whether it was written by a human or generated by AI.

Architectural thinking
The ability to understand and design system structures, make technical trade-offs and anticipate the long-term consequences of design decisions.

Hallucination (in AI)
When an AI model generates code or information that appears correct but lacks factual grounding or is technically inaccurate.

Technical debt
The consequences of short-term or insufficient technical decisions that lead to increased complexity, maintenance issues or the need for future rework.

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1. Introduction

AI is quickly transforming the landscape of software development. In recent years, AI-assisted tools – especially large language models (LLMs) – have become much better at generating code and automating tasks that used to be done by hand. Tools like GitHub Copilot, ChatGPT and Gemini can create basic code structures, suggest implementations, generate tests and even improve existing code, which means developers are increasingly working alongside AI rather than writing every line themselves.

This change brings both new opportunities and concerns. Some people think AI might fully replace developers, but others point out that AI lacks context, judgement and responsibility. While AI can make work faster, it can also cause mistakes, security problems and ethical issues if not checked by humans. The debate between excitement and caution shows why it is important to understand these changes more deeply.

This question is especially relevant for students and educators, because when AI writes much of the code, what it means to be a developer starts to change. Future skills will focus more on understanding requirements, making design decisions, collaborating in teams and using ethical judgement.

As Qiu et al. (2024) put it, AI tools are in the process of *"fundamentally altering developers' roles from manual coders to orchestrators of AI-driven development ecosystems"*. This sums up the main change: AI is not replacing developers, but it is changing their work and letting them focus on bigger-picture tasks.

It is important for both current and future developers to understand these changes. They affect how education is planned, how teams cooperate, and how companies and organizations ensure the quality and security of their systems. By looking at how AI changes developer tasks and which human skills will matter most, this report hopes to give a clearer picture of how the profession is changing and what skills will be needed in the future.

1.1 Purpose and research question

The purpose of this report is to examine how AI-assisted code generation changes and impacts the developer role, and to identify which human skills will be essential to ensure quality, ethics and correctness in AI-generated code. To address this, the following research question will be answered:

How does a developer's role change when AI takes over an increasing share of code generation, and which human skills will become essential to ensure quality, ethics and correctness in AI-generated code?

1.2 Scope

This work focuses on the developer role in modern development teams, with particular emphasis on AI-assisted code generation. The report does not address AI development in general, but specifically how the developer's tasks and competency profile are changing.

1.3 Report Structure

This section explains how the report is organized. The theory section covers key ideas, the method section explains how the study was done, the results section shares what was found, and the discussion connects these findings to the main question. The report ends with the conclusion.

2. Theory

In this section, I will explain the main ideas and viewpoints used to understand how AI changes the developer's role. The theory is split into three areas: AI-assisted code generation, human skills in modern development work, and how the developer's role is changing.

2.1 AI-assisted Code Generation

AI-assisted development refers to tools that use LLMs to generate, modify or evaluate code. These systems automate repetitive tasks and can accelerate development processes (De Visser, 2025). GitHub predicts that AI will soon generate most standard code, shifting the developer's work toward more abstract problem solving and higher-level reasoning (Davis, 2025).

LLMs work by generating the next word or line of code by using patterns from large amounts of training data. They do not understand a problem the same way humans do, instead they generate what is statistically most likely based on the data. Because of this, AI can write code that looks correct but lacks logic, contains errors, or does not fit the system's design. That is why humans need to check the results. AI can suggest new features, write tests, and improve code, but it can also hallucinate functions that do not exist or introduce security risks.

In many teams, AI is used like a "junior developer" that quickly creates drafts. The human—AI teamwork model describes exactly this: AI gives suggestions, while the human provides oversight, context and quality control (Oliver, 2025). This explains why AI can take over parts of the coding, but it cannot ensure that everything is correct or that the system is well-designed.

2.2 Human Skills in Modern Software Development

Research shows that human skills – often called soft skills or cognitive skills – are becoming increasingly important as automation grows (Cardon, 2024; Kimbrough, 2025). These skills are not about replacing technical competence but complementing it. When AI takes over more of the repetitive work, the human contribution becomes more distinct.

Architectural thinking – understanding how different parts of a system are connected and how design decisions affect the system in the long run. It requires experience and the ability to anticipate consequences – something that AI lacks, since it has no mental model of the system. A developer needs to be able to judge whether an AI suggestion fits into the bigger picture or risks creating technical debt. This is something AI cannot handle in a reliable way (Malec, 2025).

Critical reasoning – the ability to review AI-generated code, to identify errors and assess risks. Because AI often presents its suggestions with great confidence – even when they are wrong – it is important for developers to evaluate them critically rather than accepting them

automatically. Research shows that developers sometimes place too much trust in AI, which can allow errors or security issues to slip into the codebase (Oliver, 2025).

Ethical judgement – ensuring that solutions are fair, secure and follow established guidelines. This is an area where AI falls short, as it cannot reason about morality or societal impact (Marone, 2025).

Collaboration and communication – developers need to communicate with colleagues, stakeholders and users to understand needs and to make informed decisions. AI can generate code, but it cannot participate in discussions or grasp business objectives (Jaani, 2026).

Creativity and problem formulation – humans define problems, explore alternatives and think conceptually. AI can suggest changes, but it cannot determine which problem should be solved or why one solution is better than another. Creativity is therefore one of the most valuable human skills in AI-driven development environments (Oliver, 2025).

All these viewpoints show that as more technical tasks are automated, human skills matter even more. AI can write code, but people are needed to guide it, ensure quality, and give the work real meaning.

2.3 The Changing Role of the Developer

Automation shifts work roles from implementation to monitoring and design (Marone, 2025). Where much of a developer's work previously involved writing code manually, this is changing as AI takes over those tasks.

New hybrid roles are appearing, where developers review AI-generated code, write effective prompts, and ensure AI is used responsibly. These roles show that AI is not just changing what developers do, but also what skills are needed.

The division of responsibilities within a team is also shifting – senior developers take on greater responsibility for architecture and quality, while junior developers need to learn to work alongside AI from the beginning. This means that both education and workplaces need to adapt to a reality where AI is a natural part of the development process.

Sense-making theory emphasizes that humans interpret context and make ethical judgements – areas where AI lacks understanding (Marone, 2025). This is why developers still play a central role in ensuring that systems function in real-world environments and meet user needs.

Coding knowledge remains a fundamental competence. Even when AI generates code, developers need to understand it to validate correctness, maintain systems and ensure long-term quality (Plumb, 2026; Qiu et al., 2024).

2.4 Summary

To sum up, the theory shows that AI is changing what developers do, but not whether they are needed. Human skills explain why developers remain important, and theories around role shifts show how the profession is evolving. Altogether, this explains why developers will not be replaced by AI but will instead take on broader and more strategic roles.

3. Method

This section explains how the study was done. The report is based on a qualitative literature review to understand how AI-assisted code generation changes the developer's role. The work consists of three parts: selecting relevant sources, identifying common themes, and combining the results into a full picture.

3.1 Source Selection

Because this study is a literature review, sources were chosen for their relevance, credibility, and how current they are. The sources come from academic papers, industry reports, and well-known tech journalism to give a broad and solid base. Since most of the ongoing development in AI happens internationally, much of the material is in English. There are not many Swedish sources in this field, which makes international sources necessary to get a complete and current view.

3.2 Analysis of Recurring Themes

The material was studied by searching for common themes and patterns in the sources. The work was done in several steps. First, I read through all the material to get a general idea. Then I picked out key concepts and recurring arguments, like automation, human skills, risks of AI-generated code, and changing work roles.

In the next step, these concepts were grouped into broader themes. Some of the most prominent were:

- automation of coding tasks
- limitations and risks of AI-generated code
- the shift from implementation to oversight
- the growing importance of human skills such as architectural thinking, critical reasoning and ethical judgement
- the continued need for code understanding despite increased automation

3.3 Synthesis

In the synthesis, the themes were brought together to build an overall picture of how AI is changing the developer's role. Differences between academic and industry views were compared to make the analysis balanced. The aim was not just to repeat what each source said, but to see how they relate and where they agree or disagree.

This step also looked at how sources differ. Some industry reports are more positive about AI's potential, while academic sources focus more on risks and limits. By comparing these views, it becomes possible to give a more balanced picture of how things are changing.

4. Results

This section shares what was found in the literature review and what these findings mean.

4.1 AI Automates Repetitive Tasks

AI tools can generate standard code, suggest implementations and help with debugging. They can also automate routine and repetitive coding tasks. Qiu et al. (2024) describe how these tools already *"fundamentally alter developers' roles from manual coders to orchestrators of AI-driven development ecosystems",* demonstrating a clear shift away from line-by-line implementation.

4.2 Developers Are Moving Toward Architecture and Quality

As AI handles more basic coding, developers spend more time on system design, architecture, and understanding requirements (Davis, 2025; Malec, 2025). This aligns with academic views suggesting that AI augments rather than replaces human capabilities (Qiu et al., 2024).

4.3 Human Skills Are Becoming More Important

The literature agrees on this: human skills are becoming more important in AI-assisted development environments. When AI writes much of the code, developers must judge if solutions are reasonable, secure, and sustainable. This means human skills do not just add to AI – they are now essential for the development process.

Key skills highlighted in the literature include collaboration, creativity, critical reasoning, ethical judgement, adaptability, curiosity and contextual understanding.

4.4 Coding Knowledge Remains Essential

Despite increased automation, deep code understanding is still necessary. AI-generated code can contain errors, vulnerabilities or unnecessary complexity. Several sources emphasize that coding knowledge is essential for catching mistakes and ensuring long-term quality and maintainability (De Visser, 2025; Plumb, 2026).

4.5 AI Augments Developers

The sources clearly agree: AI does not replace developers but augments their capabilities. AI tools automate repetitive tasks and free up time for the more complex and creative aspects of development work (Davis, 2025; Qiu et al., 2024).

4.6 Nuances in the Source Material

Although the sources agree that human skills are becoming more important, there are nuances worth noting. Industry reports are often more optimistic, suggesting that AI will free up time for creative work, while academic sources tend to emphasize risks and the need for strict human oversight. This difference shows that the picture is not entirely straightforward – the future will likely be shaped by both opportunities and challenges.

5. Discussion

AI is handling many technical and repetitive tasks, but this also means human judgement is needed more than ever. This creates a clear paradox: the more that gets automated, the more important the parts of the work that require human insight, creativity and accountability become. The following discussion builds on these findings and connects them to earlier research and industry views.

5.1 AI Changes the Developer role But Won't Replace it

AI is effective at repetitive tasks, but being a developer also means solving problems, communicating, and designing systems. As more of the routine work is automated, developers take on responsibilities that require judgement, creativity and a broader understanding of the system. This makes the role broader and more strategic than it used to be.

Pragmatic Engineer (Orosz, 2026) offers a hopeful view, imagining a future where AI writes most of the code, but the developer's role becomes even more meaningful. Instead of focusing on syntax and implementation, developers will define problems, architecture, system behavior, and long-term sustainability. This matches the main trend in the sources: AI speeds up development, but human insight, creativity, and decision-making is still key. In the future, developers will guide AI systems instead of competing with them.

5.2 Human Skills Are Central

AI-generated code needs to be checked by humans. Developers must understand how solutions impact the system and be able to assess risks, ethics and quality. Human knowledge is needed to review, interpret and validate AI suggestions – and it goes beyond just catching bugs. It is about understanding how a solution affects the system's architecture, meets the users' needs and supports the organization's goals. Several sources also highlight that AI can introduce bias or rely on insecure dependencies, which makes ethical judgement and technical integrity crucial. The developer's role is to ensure that AI is used responsibly, not to accept its suggestions without question.

5.3 The Developer as a Creative Professional

A recurring theme in the literature is that developers are more than just coders – they are creative problem solvers. They design systems, think conceptually and iterate in ways AI cannot. Collaboration, communication and an understanding of user needs are central parts of the work. One source compares development to a jazz band: each person contributes something unique, and together they create something AI could not create alone.

In practice, this means developers use AI as a tool to try out ideas quickly, but people are the ones who drive innovation and decide the direction.

5.4 Implications for Education and Industry

The shift toward AI-assisted development means that education needs to focus on human skills just as much as technical ones. For the industry, it means that companies and organizations need to invest in skill-building, setting rules for using AI, and making sure human judgement is part of the decision-making processes.

6. Conclusion

The purpose of this report was to answer the following question:

How does a developer's role change when AI takes over an increasing share of code generation, and which human skills will become essential to ensure quality, ethics and correctness in AI-generated code?

The results show that AI is changing the developer role significantly, but not in a way that makes developers redundant. The focus is moving from writing every line of code to taking care of the bigger picture – like oversight, architecture, quality and decision-making. Developers are becoming more like orchestrators who guide and shape AI-generated solutions, instead of just writing code themselves.

Human skills – such as architectural thinking, critical reasoning, creativity, collaboration, ethical judgement and coding knowledge – are central to this shift. These are the skills that help developers understand business needs, interpret context, identify risks and make decisions that AI cannot. They balance the limitations of AI systems and ensure that the technology is used in a responsible and sustainable way.

As Pragmatic Engineer notes, the future of the industry is not about people disappearing – it is about AI taking over the mechanical side of coding while developers focus on creativity, architecture and shaping the systems that AI helps to build.

Overall, the findings suggest that the developer's role will be broader and more strategic in the future. The role is less about writing every line of code and more about understanding the big picture, making trade-offs, communicating with stakeholders and combining human insight with AI-generated solutions. Developers will also need to stay curious, adapt, and keep learning as technology changes.

Finally, this report shows that AI will not reduce the need for people – it will change which parts of the development work require human expertise. As Davis (2025) puts it: *"Robots are terrible at creativity, collaboration, and big-picture thinking. Which, conveniently, is where the future of development is headed."* And it is precisely in those areas that developers will continue to be indispensable.

References

Cardon, P. (2024). New study finds AI is making soft skills more important in the workplace. *World Economic Forum*. Retrieved February 11, 2026, from https://www.weforum.org/stories/2024/01/ai-value-soft-skills-workplace-jobs/

Davis, G. (2025). The developer role is evolving: Here's how to stay ahead. *GitHub Blog*. Retrieved February 11, 2026, from https://github.blog/ai-and-ml/the-developer-role-is-evolving-heres-how-to-stay-ahead/

De Visser, H. (2025). The future of application development in the AI era. *DevOps.com*. Retrieved February 11, 2026, from https://devops.com/the-future-of-application-development-in-the-ai-era/

Jaani, J. (2026). Mänskliga färdigheter skapar framtidens arbetssätt. [PDF]. Retrieved February 11, 2026, from https://digitalatips.se/wp-content/uploads/2026/01/Manskliga-fardigheter-skapar-framtidens-arbetssatt_it.pdf

Kimbrough, K. (2025). AI is shifting the workplace skillset. But human skills still count. *World Economic Forum*. Retrieved February 11, 2026, from https://www.weforum.org/stories/2025/01/ai-workplace-skills/

Malec, M. (2025). The AI development team of the future: Skills, roles & structure. *HatchWorks*. Retrieved February 11, 2026, from https://hatchworks.com/blog/gendd/ai-development-team-of-the-future/

Marone, M. (2025). The fluid future of work: Rethinking roles in the age of intelligent machines. *Harvard Business Publishing*. Retrieved February 11, 2026, from https://www.harvardbusiness.org/insight/the-fluid-future-of-work-rethinking-roles-in-the-age-of-intelligent-machines/

Oliver, D. R. (2025). The evolution of software development in the age of AI: Will developers become obsolete? *Medium*. Retrieved February 11, 2026, from https://medium.com/@davidroliver/the-evolution-of-software-development-in-the-age-of-ai-will-developers-become-obsolete-519ab57526bb

Plumb, R. (2026). Prognos: AI kommer inte att ersätta mänskliga utvecklare på minst fem år. *Computer Sweden*. Retrieved February 11, 2026, from https://computersweden.se/article/4113662/prognos-ai-kommer-inte-att-ersatta-manskliga-utvecklare-pa-minst-fem-ar.html

Qiu, K., Puccinelli, N., Ciniselli, M., & Di Grazia, L. (2024). From today's code to tomorrow's symphony: The AI transformation of developers' routine by 2030. *ACM Transactions on Software Engineering and Methodology*. Retrieved February 11, 2026, from [Preprint]. arXiv. https://arxiv.org/abs/2405.12731

Orosz, G. (2026). When AI writes almost all code — what happens to developers? *Pragmatic Engineer*. Retrieved February 18, 2026, from https://newsletter.pragmaticengineer.com/p/when-ai-writes-almost-all-code-what